

Sci-kit Learn Recreation and Implementation

CSE299-sec-5

Group-5

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Week 5 ML Library Implementation - Project Update

Key Accomplishments

Tree-Based Models Implementation: Successfully developed Decision Tree and Random Forest algorithms with comprehensive feature sets including configurable hyperparameters such as tree depth, splitting criteria (Gini and entropy), minimum samples per split/leaf, and bootstrap aggregating for ensemble methods. Decision Tree implementation includes recursive tree building, feature importance calculation, and pruning capabilities, while Random Forest provides ensemble voting, out-of-bag scoring, and random feature selection at each split.

Advanced Model Selection Framework: Implemented robust cross-validation system supporting both k-fold and stratified splitting strategies with detailed performance metrics, timing analysis, and individual fold results reporting. Developed comprehensive hyperparameter tuning capabilities through Grid Search and Randomized Search utilities, each integrated with cross-validation and supporting multiple scoring metrics including accuracy, precision, recall, and F1-score. **Enhanced CLI Integration:** Extended command-line interface with tree-specific parameters (max_depth, min_samples_split, criterion, n_estimators) and validation options (cross_validate, grid_search, randomized_search, stratified), enabling systematic model comparison and reproducible experimentation workflows. **Professional**

Validation Infrastructure: Created comprehensive model evaluation system with automatic feature importance reporting, tree-specific metrics (depth, leaves, OOB scores), and detailed cross-validation results including confidence intervals and timing benchmarks. The framework now supports rigorous model selection workflows with systematic parameter exploration and performance comparison across different algorithms and datasets. **Quality Assurance and Testing:** Conducted extensive testing across multiple datasets (Iris, Wine, Digits) with various model configurations, ensuring robust performance and reliable results. Implemented parameter validation utilities and error handling for edge cases, maintaining code quality and user experience.

